
Gaël-Pacôme Nguimeya Tematio

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PROFILE

I am a researcher in computational condensed matter and materials physics with expertise in Density Functional Theory (DFT), electronic structure calculations, atomistic materials modeling, and topological materials. My experience in first-principles simulations, tight-binding modeling, Python-based workflow automation, and high-performance computing enables me to contribute to theoretical and computational studies of nonmagnetic, magnetic, and topological quantum materials. My broader research interests include quantum physics, quantum simulation, and the use of data-driven methods to support materials discovery.

TECHNICAL SKILLS

DFT & Atomistic Materials Modeling: Proficient in Quantum Espresso, Wannier90, WannierTools, and Quantum Lattice for first-principles simulations, electronic-structure calculations, tight-binding modeling, and analysis of structure-property relationships in materials.

Crystallography & Visualization: Skilled in Xcrysden, FermiSurfer, Vesta, and Avogadro for molecular and crystal structure analysis, visualization, and interpretation of computational materials data.

Scientific Programming & Workflow Automation: Advanced knowledge of Python and Bash scripting for automating simulations, managing computational workflows, performing data analysis, and supporting high-throughput materials modeling.

Data-Driven & Quantum Computing Methods: Working exposure to machine learning, applied data science, and quantum machine learning through recent training; experienced with Qiskit, PennyLane, Classiq, and Cirq.

Version Control, Documentation & Operating Systems: Proficient in Git, GitHub, Google Apps Script, and LaTeX (including Beamer); full competency in Linux and Windows environments, with experience using Unix-based computational workflows.

EDUCATION & TRAINING

Ph.D., Computational Condensed Matter and Materials Physics, University of Cape Town, Sept 2025
South Africa

Research Master, Condensed Matter Physics, University of Dschang, Cameroon Sept 2015

Structured Master, Mathematical Sciences, the African Institute for Mathematical Sciences (AIMS) Cameroon Jun 2015

Bachelor of Science, Condensed Matter Physics, University of Dschang, Cameroon Sept 2012

LIST OF PUBLICATIONS

- [3] Gaël-Pacôme Nguimeya Tematio, Trisha Salagaram, Aniekan Magnus Ukpog, Houssam Eddine Hailouf, and Kingsley Onyebuchi Obodo. Strain and electric field control of topological phases in $SrSi_2$ and $CoSi$: A first-principles and model Hamiltonian study. *Mater. Chem. Phys.* 132574 (2026). <https://doi.org/10.1016/j.matchemphys.2026.132574>
- [2] Gaël-Pacôme Nguimeya Tematio, Trisha Salagaram, Aniekan Magnus Ukpog, Houssam Eddine Hailouf, and Kingsley Onyebuchi Obodo. Field-driven electronic and topological modulation in the Weyl semimetal NbP. *J. Phys.: Condens. Matter* **38**, 115501 (2026). <https://doi.org/10.1088/1361-648X/ae50a2>

- [1] Kenfack SC, Nguimeya GP, Talla PK, Fotue AJ, Fobasso MFC, Tiotso M, Djomou JRD, Ekosso CM and Fai LC. Decoherence of driven coupled harmonic oscillator. *J. Nanosci. Curr. Res.* **1**, 1000104 (2016). <https://doi.org/10.4172/2572-0813.1000104>

TEACHING

Teaching assistant (Tutor), African Institute for Mathematical Sciences (AIMS), Ghana	<i>Mar 2025 - Jun 2025</i>
Teaching assistant (Head Tutor), African Institute for Mathematical Sciences (AIMS), South Africa	<i>Sept 2022 - Oct 2023</i>
Teaching assistant (Tutor), African Institute for Mathematical Sciences (AIMS), South Africa	<i>Feb 2022 - Jun 2022</i>
Teaching assistant, University of Cape Town (UCT), South Africa	<i>Feb 2019 - Nov 2019</i>
Teaching assistant, Higher Teacher Training College (HTTC) Bambili, Cameroon.	<i>Feb 2017 - Mar 2017</i>
• Quantum Mechanics (Undergraduate course). • Advanced Quantum Mechanics (Graduate course).	
Part-time High School Physics Teacher, Government Bilingual High School (GBHS) Mbouda, Cameroon	<i>Sept 2016 - Jun 2017</i>
Teaching assistant, University of Dschang, Cameroon.	<i>May 2016 - Jun 2016</i>
• Quantum mechanics (Graduate course).	
Teaching assistant, Higher Teacher Training College (HTTC) Bambili, Cameroon.	<i>Feb 2016 - Mar 2016</i>
• Quantum mechanics (Graduate course).	
Teaching assistant, University of Dschang, Cameroon.	<i>Nov 2015 - Jan 2016</i>
• Electromagnetism (Undergraduate course). • Electrostatics (Undergraduate course). • General Physics (Undergraduate course).	
Teaching assistant, University of Dschang, Cameroon.	<i>May 2014 - Jul 2014</i>
• Introduction to Quantum Physics (Undergraduate course).	

CERTIFICATIONS

Badge Womanium 2025 Quantum Program issued by The Washington Institute for STEM, Entrepreneurship and Research.	<i>Aug 2025</i>
Quantum Algorithms to Solve Partial Differential Equations issued by The Washington Institute for STEM, Entrepreneurship and Research.	<i>Aug 2025</i>
Quantum Algorithms for Nonlinear Problems issued by The Washington Institute for STEM, Entrepreneurship and Research.	<i>Aug 2025</i>
PennyLane Trotterization Challenge at the Womanium & Wiser Quantum Program 2025 issued by PennyLane.	<i>Aug 2025</i>
Qiskit Global Summer School 2025 - Quantum Excellence Summer School by IBM Quantum.	<i>Aug 2025</i>
PennyLane Linear Combination of Unitaries (LCU) Challenge at the Womanium & Wiser Quantum Program 2025 issued by PennyLane.	<i>Jul 2025</i>

Quantathon (Hackathon at AIMS Ghana) by AIMS Ghana QTedu Global Event:
a week-long celebration of innovation, collaboration, and quantum education for
the SDGs; awarded second place to my team.

Jul 2025

Full list on LinkedIn: <https://www.linkedin.com/in/pacomenguimeya>.

LEADERSHIP ROLES

Class Representative: Class representative for the MSc in Condensed
Matter Physics at the University of Dschang, Cameroon.

Oct 2013 - Sept 2015

LANGUAGES

French: Mother tongue

English: Fluent

REFERENCES

Available upon request.